

2022-2023 Fall
MAT123 Final
13/01/2023

1. Evaluate the following definite integrals.

a. $\int_0^1 \frac{x^2}{\sqrt{4-x^2}} dx$

b. $\int_0^{\pi/4} x \sec^2 x dx$

2. a. Consider the finite region between the curves $y = e^{-x}$, $y = x/e$, and the y -axis. Write a definite integral (do not evaluate) by using the Cylindrical Shell Method, which gives the volume of the solid obtained by rotating this region about the x -axis.
- b. Consider the infinite region between the curves $y = e^{-x}$, $y = x/e$ and the x -axis. Write a definite integral (do not evaluate) by using the Washer Method, which gives the volume of the solid obtained by rotating this region about the line $y = -1$.
- c. Evaluate the area of the region given in part (a).

3. Determine whether each series is convergent or divergent.

a. $\sum_{n=1}^{\infty} \frac{1}{\arctan(n^2)}$

b. $\sum_{n=1}^{\infty} \sin\left(\frac{1}{n^3}\right)$

c. $\sum_{n=1}^{\infty} n e^{-n^2}$

4. a. Find the radius and interval of convergence of the power series

$$\sum_{n=1}^{\infty} \frac{2^{n+1} \cdot (x+1)^n}{n \cdot 3^n}.$$

- b. Using the formula $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$ for $|x| < 1$, find the Maclaurin series of the function

$$f(x) = \frac{x^{123}}{1+x^4}.$$

1. a. Let $x = 2 \sin u$ for $-\frac{\pi}{2} < u < \frac{\pi}{2}$, then $dx = 2 \cos u \, du$.

$$x = 0 \implies 2 \sin u = 0 \implies u = 0$$

$$x = 1 \implies 2 \sin u = 1 \implies \sin u = \frac{1}{2} \implies u = \arcsin \frac{1}{2} = \frac{\pi}{6}$$

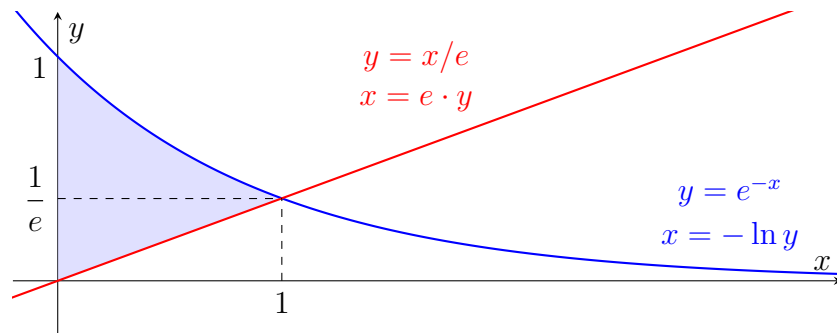
$$\begin{aligned} I &= \int_0^1 \frac{x^2}{\sqrt{4-x^2}} dx = \int_0^{\pi/6} \frac{4 \sin^2 u}{\sqrt{4-4 \sin^2 u}} \cdot 2 \cos u \, du = \int_0^{\pi/6} \frac{4 \sin^2 u \cos u}{|\cos u|} du \\ &= \int_0^{\pi/6} 4 \sin^2 u \, du = 4 \int_0^{\pi/6} (1 - \cos^2 u) \, du = 4 \int_0^{\pi/6} \frac{1 - \cos 2u}{2} du \\ &= 4 \left(\frac{u}{2} - \frac{\sin 2u}{4} \right) \Big|_0^{\pi/6} = 2u - \sin 2u \Big|_0^{\pi/6} = \left(\frac{\pi}{3} - \sin \frac{\pi}{3} \right) - 0 = \boxed{\frac{\pi}{3} - \frac{\sqrt{3}}{2}} \end{aligned}$$

- b. Use the method of integration by parts.

$$\left. \begin{array}{l} u = x \implies du = dx \\ dv = \sec^2 x \, dx \implies v = \tan x \end{array} \right\} \rightarrow \int_a^b u \, dv = uv \Big|_a^b - \int_a^b v \, du$$

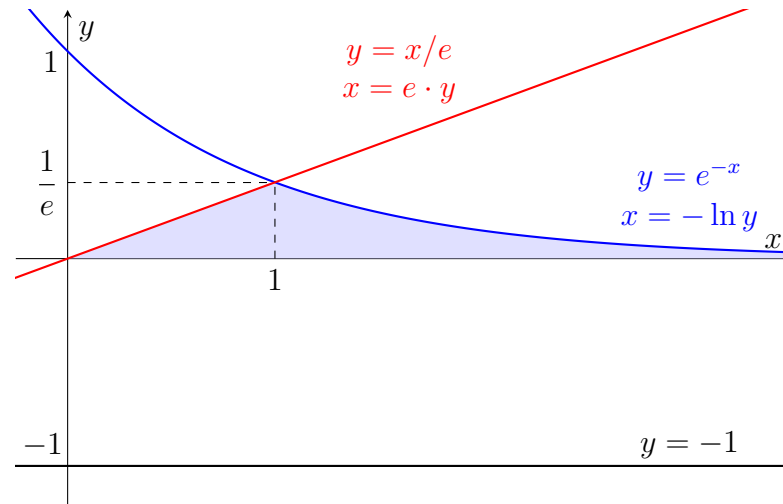
$$\begin{aligned} \int_0^{\pi/4} x \sec^2 x \, dx &= x \tan x \Big|_0^{\pi/4} - \int_0^{\pi/4} \tan x \, dx = x \tan x + \ln |\cos x| \Big|_0^{\pi/4} \\ &= \boxed{\frac{\pi}{4} + \ln \frac{\sqrt{2}}{2}} \end{aligned}$$

2. a.



$$V = \int_D 2\pi h(y) r(y) \, dy = \boxed{\int_0^{1/e} 2\pi y (ey - 0) \, dy + \int_{1/e}^1 2\pi y (-\ln y - 0) \, dy}$$

b.



$$V = \int_D \pi [r_2^2(x) - r_1^2(x)] dx = \boxed{\int_0^1 \left[\left(\frac{x}{e} + 1 \right)^2 - 1^2 \right] dx + \int_1^\infty \left[(e^{-x} + 1)^2 - 1^2 \right] dx}$$

c. The volume can be expressed by the following integral.

$$V = 2\pi \left[\int_0^{1/e} ey^2 dy - \int_{1/e}^1 y \ln y dy \right] \quad (1)$$

Calculate the integral on the left in (??).

$$\int_0^{1/e} ey^2 dy = \frac{ey^3}{3} \Big|_0^{1/e} = \frac{1}{3e^2}$$

Calculate the integral on the right in (??) by using the method of integration by parts.

$$\left. \begin{array}{l} u = \ln x \implies du = \frac{1}{x} dx \\ dv = x dx \implies v = \frac{x^2}{2} \end{array} \right\} \rightarrow \int_a^b u dv = uv \Big|_a^b - \int_a^b v du$$

$$\begin{aligned} \int_{1/e}^1 x \ln x dx &= \frac{x^2 \ln x}{2} \Big|_{1/e}^1 - \int_{1/e}^1 \frac{x^2}{2} \cdot \frac{1}{x} dx = \frac{x^2 \ln x}{2} - \frac{x^2}{4} \Big|_{1/e}^1 \\ &= \left(0 - \frac{1}{4} \right) - \left(-\frac{1}{2e^2} - \frac{1}{4e^2} \right) = \frac{3}{4e^2} - \frac{1}{4} \end{aligned}$$

Therefore,

$$V = 2\pi \left[\left(\frac{1}{3e^2} \right) - \left(\frac{3}{4e^2} - \frac{1}{4} \right) \right] = \boxed{\pi \left(\frac{1}{2} - \frac{5}{6e^2} \right)}.$$

3. Determine whether each series is convergent or divergent.

- a.** Apply the n th Term Test for divergence. The inverse trigonometric function $\arctan x$ is continuous on $\mathbb{R}$. Therefore, we may take the limit inside the function.

$$\lim_{n \rightarrow \infty} \frac{1}{\arctan(n^2)} = \frac{1}{\arctan\left(\lim_{n \rightarrow \infty} n^2\right)} = \frac{1}{\frac{\pi}{2}} = \frac{2}{\pi} \neq 1$$

By the n th Term Test for divergence, the series $\sum_{n=1}^{\infty} \frac{1}{\arctan(n^2)}$ diverges.

- b.** Recall the sine inequality $-\theta \leq \sin \theta \leq \theta$. Then for all $n \in \mathbb{R}$ except zero, we have $-\frac{1}{n^3} \leq \sin\left(\frac{1}{n^3}\right) \leq \frac{1}{n^3}$. The series $\sum_{n=1}^{\infty} \frac{1}{n^3}$ converges because it is a p -series with $p = 3 > 1$. By the p -series Test, the series converges. The series $\sum_{n=1}^{\infty} \sin\left(\frac{1}{n^3}\right)$ also converges by the Direct Comparison Test because $\sin\left(\frac{1}{n^3}\right) < \frac{1}{n^3}$ for every $n \geq 1$.
- c.** Take $f(x) = xe^{-x^2}$. f is continuous because the product of a polynomial and an exponential expression is still continuous. f is positive and decreasing for $x \geq 1$. Verify the monotonicity of f by taking the first derivative.

$$\frac{df}{dx} = 1 \cdot e^{-x^2} + xe^{-x^2} \cdot (-2x) = e^{-x^2} (1 - 2x^2)$$

$$f'(x) < 0 \quad \text{for} \quad x > \frac{\sqrt{2}}{2} \implies f'(x) < 0 \quad \text{for} \quad x \geq 1$$

We may now apply the Integral Test. Take the limit for the improper integral.

$$\int_1^{\infty} xe^{-x^2} dx = \lim_{R \rightarrow \infty} \int_1^R xe^{-x^2} dx = \lim_{R \rightarrow \infty} \left. -\frac{1}{2}e^{-x^2} \right|_1^R = \lim_{R \rightarrow \infty} -\frac{1}{2} \left(e^{-R^2} - e^{-1} \right) = \frac{1}{2e}$$

The integral converges. Then the series $\sum_{n=1}^{\infty} ne^{-n^2}$ also converges.

- 4. a.** Apply the Ratio Test for absolute convergence, and apply other tests at the endpoints.

$$\begin{aligned} \lim_{n \rightarrow \infty} \left| \frac{2^{n+2} \cdot (x+1)^{n+1}}{(n+1) \cdot 3^{n+1}} \cdot \frac{n \cdot 3^n}{(x+1)^n \cdot 2^{n+1}} \right| &= \lim_{n \rightarrow \infty} \left| \frac{2n \cdot (x+1)}{(n+1) \cdot 3} \right| \\ &= \frac{2|x+1|}{3} \cdot \lim_{n \rightarrow \infty} \left| \frac{n}{n+1} \right| = \frac{2|x+1|}{3} \end{aligned}$$

$$\frac{2|x+1|}{3} < 1 \implies |x+1| < \frac{3}{2}$$

The radius of convergence is $\boxed{\frac{3}{2}}$.

$$|x+1| < \frac{3}{2} \implies -\frac{3}{2} < x+1 < \frac{3}{2} \implies -\frac{5}{2} < x < \frac{1}{2} \quad (\text{convergent})$$

Investigate the convergence at the endpoints.

$$x = \frac{1}{2} \implies \sum_{n=1}^{\infty} \frac{2^{n+1} \cdot \left(\frac{3}{2}\right)^n}{n \cdot 3^n} = \sum_{n=1}^{\infty} \frac{2}{n} = 2 \cdot \sum_{n=1}^{\infty} \frac{1}{n}$$

This is a p -series with $p = 1$, for which the series diverges by the p -series Test. Try the other endpoint.

$$x = -\frac{5}{2} \implies \sum_{n=1}^{\infty} \frac{2^{n+1} \cdot \left(-\frac{3}{2}\right)^n}{n \cdot 3^n} = \sum_{n=1}^{\infty} \frac{2 \cdot (-1)^n}{n} = 2 \cdot \sum_{n=1}^{\infty} \frac{(-1)^n}{n}$$

This is an alternating series. The non-alternating part, which is $\frac{1}{n}$, is nonincreasing for $n \geq 1$ and it is positive. The limit at infinity is 0. By Leibniz's Alternating Series Test, the series converges.

The convergence set for the power series is $\boxed{\left[-\frac{5}{2}, \frac{1}{2}\right)}$.

b. Rewrite the function.

$$f(x) = \frac{x^{123}}{1+x^4} = x^{123} \cdot \frac{1}{1-(-x^4)} = x^{123} \cdot \sum_{n=0}^{\infty} (-x^4)^n = x^{123} \cdot \sum_{n=0}^{\infty} (-1)^n \cdot x^{4n}$$

$$= \boxed{\sum_{n=0}^{\infty} (-1)^n \cdot x^{123+4n} = x^{123} - x^{127} + x^{131} - x^{135} + \dots}$$